Aquifer, reservoir, plant

Purpose

This page covers four groups of output files:

- aquifer storage and fluxes (`aquifer_*.txt`, `basin_aqu_*.txt`)
- reservoir storage and fluxes (`reservoir_*.txt`, `basin_res_*.txt`)
- crop yields (`crop_yld_*.txt`, `basin_crop_yld_*.txt`)
- plant growth, stress and weather (`hru_pw_*.txt`, `hru-lte_pw_*.txt`, `lsunit_pw_*.txt`, `basin_pw_*.txt`)

Each file is switched on in `print.prt`. The suffix gives the print interval: `_day`, `_mon`, `_yr` or `_aa` (average annual). When `csvout` is `y`, a `.csv` copy with the same columns is written beside each `.txt` file. The first seven columns of every file except the crop yield files are `jday`, `mon`, `day`, `yr`, `unit`, `gis_id` and `name`; see Naming convention.

Aquifer

Files and switches

File	`print.prt` row	Scope
`aquifer_day.txt`, `aquifer_mon.txt`, `aquifer_yr.txt`, `aquifer_aa.txt`	`aquifer`	one row per aquifer per interval
`basin_aqu_day.txt`, `basin_aqu_mon.txt`, `basin_aqu_yr.txt`, `basin_aqu_aa.txt`	`basin_aqu`	one row per interval for the whole basin

The aquifer files report the one-dimensional aquifers defined in `aquifer.aqu` and are opened only when the model has such aquifers. When the gwflow module replaces the aquifers, use the gwflow outputs instead.

Columns

Line 1 is the project name and version, line 2 the column header and line 3 the units. After the seven leading columns come 17 value columns:

#	Column	Units	Description
1	`flo`	mm	return flow leaving the aquifer
2	`dep_wt`	m	depth from the ground surface to the water table
3	`stor`	mm	water stored in the aquifer
4	`rchrg`	mm	recharge entering the aquifer from other objects
5	`seep`	mm	deep seepage from the bottom of the aquifer
6	`revap`	mm	water removed by plant uptake and evaporation (revap)
7	`no3_st`	kg N/ha	nitrate-N stored in the aquifer
8	`minp`	kg P	mineral P sent out with the return flow (see the notes)
9	`orgn` (header)	%	organic carbon content of the aquifer (see the notes)
10	`orgp` (header)	kg N/ha	organic N in the aquifer (see the notes)
11	`no3_rchg`	kg N/ha	nitrate-N entering with recharge; `rchrgn` in earlier documentation

#	Column	Units	Description
12	`no3_loss`	kg N/ha	nitrate-N lost in the aquifer (decay); `nloss` in earlier documentation
13	`no3_lat`	kg N/ha	nitrate-N leaving with the return flow; `no3gw` in earlier documentation
14	`no3_seep`	kg N/ha	nitrate-N leaving with deep seepage; `seepno3` in earlier documentation
15	`flo_cha`	mm	part of `flo` routed to channels
16	`flo_res`	mm	part of `flo` routed to reservoirs
17	`flo_ls`	mm	part of `flo` routed to other aquifers

Notes on individual columns:

- Columns 9 and 10 are mislabelled. The header says `orgn` (kg/ha_N) and `orgp` (kg/ha_P), but column 9 is the organic carbon content from the `carbon` column of `aquifer.aqu`, constant through the run, and column 10 is organic N, which 62.0.1 never updates, so it stays zero.
- `minp` is the mineral P concentration from the `sol_p` column of `aquifer.aqu` multiplied by the day's recharge volume. The value is a mass in kg for the whole aquifer; the units line gives kg/ha_P. It is computed from the recharge volume and leaves the aquifer with the return flow.
- `flo_cha`, `flo_res` and `flo_ls` split `flo` by the outflow fractions in the aquifer's connection file. Only outflows of type `tot` are counted, so the three columns add up to `flo` only when every outflow is of that type.

Aggregation

Fluxes are summed over the interval. Three state columns are averaged instead: the monthly file divides `stor`, `dep_wt` and `no3_st` by the number of days in the month, and the yearly file divides the sum of the monthly averages by 12. The average annual file divides the yearly values by the number of printed years. Column 9 is never summed: every interval reports the starting value.

The basin file is the area-weighted average of all aquifers, each weighted by its share of the basin area.

Example header

`basin_aqu_yr.txt` from the `Ames_sub1` reference dataset run with 62.0.1:

```
   jday   mon   day    yr   unit  gis_id  name                          flo          dep_wt             stor
                                                                         mm               m               mm
```

Reservoir

Files and switches

File	`print.prt` row	Scope
`reservoir_day.txt`, `reservoir_mon.txt`, `reservoir_yr.txt`, `reservoir_aa.txt`	`reservoir`	one row per reservoir per interval
`basin_res_day.txt`, `basin_res_mon.txt`, `basin_res_yr.txt`, `basin_res_aa.txt`	`basin_res`	all reservoirs added together

The reservoir files are opened only when the model has reservoirs.

Columns

After the seven leading columns come four blocks: 4 water body columns, then 18 storage, 18 inflow and 18 outflow columns.

Block	Columns	Units
Water body	`area`, `precip`, `evap`, `seep`	ha, m³, m³, m³
Storage at the end of the interval	`flo_stor`, `sed_stor`, `orgn_stor`, `sedp_stor`, `no3_stor`, `solp_stor`, `chla_stor`, `nh3_stor`, `no2_stor`, `cbod_stor`, `dox_stor`, `san_stor`, `sil_stor`, `cla_stor`, `sag_stor`, `lag_stor`, `grv_stor`, `null`	see below
Inflow during the interval	the same 18 names ending in `_in`	see below
Outflow during the interval	the same 18 names ending in `_out`	see below

Units of the 18 constituent columns, which repeat in the storage, inflow and outflow blocks:

Constituent	Units	Description
`flo`	m³	water volume
`sed`	t	sediment
`orgn`	kg N	organic N
`sedp`	kg P	organic (sediment-attached) P
`no3`	kg N	nitrate-N
`solp`	kg P	soluble mineral P
`chla`	kg	chlorophyll-a
`nh3`	kg N	ammonium-N
`no2`	kg N	nitrite-N
`cbod`	kg	carbonaceous biochemical oxygen demand
`dox`	kg	dissolved oxygen
`san`, `sil`, `cla`, `sag`, `lag`, `grv`	t	sand, silt, clay, small aggregates, large aggregates and gravel
`null`	°C	water temperature (the header gives `null` and no unit)

The inflow columns are `flo_in`, `sed_in`, `orgn_in`, `sedp_in`, `no3_in`, `solp_in`, `chla_in`, `nh3_in`, `no2_in`, `cbod_in`, `dox_in`, `san_in`, `sil_in`, `cla_in`, `sag_in`, `lag_in`, `grv_in` and `null`. The outflow columns are `flo_out`, `sed_out`, `orgn_out`, `sedp_out`, `no3_out`, `solp_out`, `chla_out`, `nh3_out`, `no2_out`, `cbod_out`, `dox_out`, `san_out`, `sil_out`, `cla_out`, `sag_out`, `lag_out`, `grv_out` and `null` (the last column of each block is headed `null`).

Inflow and outflow are volumes and masses for the interval, in m³, t and kg. The storage block is the reservoir content when the line is written, so in the monthly, yearly and average annual files it is the storage on the last day of the interval.

Aggregation

Inflow, outflow, precipitation, evaporation and seepage are summed over the interval, and the average annual file divides the totals by the number of printed years. The surface `area` is averaged over the days of the month, and the yearly value is the average of the monthly values.

`reservoir_mon.csv` has one extra line after the units line, `RES  reservoir_mon.csv`, which the other reservoir files do not have.

Crop yield

Files and switches

The `crop_yld` column of `print.prt` controls all crop yield files: `y` gives yearly files, `a` average annual files, `b` both, and `n` none.

File	Written when `crop_yld` is	Content
`crop_yld_aa.txt`	`a` or `b`	one row per HRU and plant at the end of the run

File	Written when `crop_yld` is	Content
`crop_yld_yr.txt`	`y` or `b`	header lines only (see below)
`basin_crop_yld_yr.txt`	`y` or `b`	one row per plant per year for the whole basin
`basin_crop_yld_aa.txt`	`y` or `b`	one row per plant at the end of the run

With `crop_yld = a`, only `crop_yld_aa.txt` is produced; the basin files need `y` or `b`. In 62.0.1 no data rows are written to `crop_yld_yr.txt` (or its `.csv`). For yearly yields use `basin_crop_yld_yr.txt`, or `yield.out` (switched on by `mgtout`) for every harvest.

HRU columns

`crop_yld_aa.txt` has a title line, a line `--YIELD (kg/ha)--`, the column header, and one row per plant in each HRU:

#	Column	Units	Description
1	`jday`	-	day of year of the last simulated day
2	`mon`	-	month
3	`day`	-	day of month
4	`yr`	-	year
5	`unit`	-	HRU number
6	`PLANTNM`	-	plant name from `plants.plt`
7	`MASS`	kg/ha	dry mass of the harvested yield
8	`C`	kg/ha	carbon in the yield
9	`N`	kg/ha	nitrogen in the yield
10	`P`	kg/ha	phosphorus in the yield

The values are the total yield over the whole run divided by the number of harvests of that plant, that is, the average yield per harvest. A plant that was never harvested reports zero. The `.csv` header lists a single `yield` column, while each data row contains the four values `MASS`, `C`, `N` and `P`.

Basin columns

#	Column	Units	Description
1	`year`	-	calendar year (the last year in the `_aa` file)
2	`plant_no`	-	plant number in the basin plant list
3	`plant_name`	-	plant name
4	`harv_area(ha)`	ha	area harvested; an HRU harvested twice in a year counts twice
5	`yld(t)`	t	total dry yield
6	`yld(t/ha)`	t/ha	`yld(t)` divided by `harv_area(ha)`

The yearly file skips the years before `nyskip`. In the average annual file, area and yield are divided by the number of printed years, and `yld(t/ha)` is total yield divided by total harvested area.

Example

`crop_yld_aa.txt` and `basin_crop_yld_aa.txt` from a two-year run of the `Ames_sub1` reference dataset with 62.0.1:

```
                                                                --YIELD (kg/ha)--
  jday    mon    day     yr    unit PLANTNM                MASS          C           N           P
   366     12     31   1976       1    corn                             3692.158    1661.471
   366     12     31   1976       1    soyb                                0.000       0.000
```

```
 year      plant_no   plant_name       harv_area(ha)     yld(t)          yld(t/ha)
  1976            1 corn                                   12.04350       47.48362       3.942675
  1976            2 soyb                                0.0000000E+00  0.0000000E+00  0.0000000
```

Plant and weather

Files and switches

File	print.prt row	Scope
`hru_pw_day.txt` to `hru_pw_aa.txt`	`hru_pw`	one row per HRU
`hru-lte_pw_day.txt` to `hru-lte_pw_aa.txt`	`hru-lte_pw`	one row per HRU-lte
`lsunit_pw_day.txt` to `lsunit_pw_aa.txt`	`lsunit_pw`	one row per landscape unit
`basin_pw_day.txt` to `basin_pw_aa.txt`	`basin_pw`	whole basin

All use the columns below. The HRU files add `plant_cov` and `mgt_ops` at the end of each row: the plant community and the management schedule of the HRU's land use.

Columns

#	Column	Units	Description
1	`lai`	m²/m²	leaf area index
2	`bioms`	kg/ha	total plant biomass
3	`yield`	kg/ha	harvested yield (dry mass)
4	`residue`	kg/ha	surface residue
5	`sol_tmp`	°C	temperature of soil layer 2
6	`strsw`	days	water (drought) stress days
7	`strsa`	days	aeration (excess water) stress days
8	`strstmp`	days	temperature stress days
9	`strsn`	days	nitrogen stress days
10	`strsp`	days	phosphorus stress days
11	`strss`	days	salinity stress days
12	`nplt`	kg N/ha	nitrogen in the plant biomass
13	`percn`	kg N/ha	nitrate-N leached below the soil profile
14	`pplnt`	kg P/ha	phosphorus in the plant biomass
15	`tmx`	°C	maximum air temperature
16	`tmn`	°C	minimum air temperature
17	`tmpav`	°C	mean air temperature
18	`solarad`	MJ/m²	solar radiation
19	`wndspd`	m/s	wind speed
20	`rhum`	fraction	relative humidity
21	`phubas0`	°C	accumulated base-zero heat units
22	`lai_max`	m²/m²	maximum leaf area index (always 0 in 62.0.1)
23	`bm_max`	kg/ha	maximum total biomass during the interval
24	`bm_grow`	kg/ha	biomass growth
25	`c_gro`	kg C/ha	plant carbon growth

The units line writes `kgha` for columns 2 to 4 and 12 to 14 and `----` for the stress columns.

Aggregation

Columns	Monthly	Yearly	Average annual
`yield`, stress days, `percn`	sum	sum	yearly sum averaged over the printed years

Columns	Monthly	Yearly	Average annual
nplt, pplnt	plant N and P at the end of the month	at the end of the year	at the end of the run
bm_max	maximum in the month	maximum in the year	maximum in the run
all other columns	daily average	daily average	daily average

nplt and pplnt are the N and P contents of the standing plants. They do not report uptake during the interval. After a harvest or kill they drop to zero, so yearly and average annual values are often zero for annual crops.

Example

hru_pw_mon.txt, Ames_sub1 run with 62.0.1 (header, units and first data row):

jday	mon	day	yr	unit	gis_id	name	lai	bioms	yield	residue
							m**2/m**2	kgha	kgha	kgha
31	1	31	1975	1	1	hru0001	0.000	0.000	0.000	17278.244

Related files

- print.prt: switches for every file on this page.
- aquifer.aqu and reservoir.res: the objects reported here.
- Water balance outputs and Channel and sediment outputs.
- gwflow outputs: groundwater output when gwflow replaces the aquifers.

Sources

- Aquifer: aquifer_output.f90 (units 2520 to 2527), basin_aquifer_output.f90 (units 2090 to 2097), headers and arithmetic in aquifer_module.f90, file opening in header_aquifer.f90 and header_write.f90, values set in aqu_1d_control.f90 and aqu_initial.f90
- Reservoir: reservoir_output.f90 (units 2540 to 2547), basin_reservoir_output.f90, header_reservoir.f90, water body type in water_body_module.f90, constituent headers in hydrograph_module.f90
- Crop yield: hru_output.f90 (unit 4008), file opening in output_landscape_init.f90 (units 4008 to 4011) and header_yield.f90, basin rows in time_control.f90 (units 5100 and 5101)
- Plant and weather: hru_output.f90 (units 2040 to 2047), daily values in hru_control.f90, type, header and operators in output_landscape_module.f90
- print.prt rows: basin_print_codes_read.f90

Output column	Fortran variable
aquifer column 9 (orgn in the header)	aqu_d%cbn (from aqu_dat%cbn)
aquifer column 10 (orgp in the header)	aqu_d%orgn
aquifer minp	aqu_d%minp
reservoir water body block	res_wat_d (type water_body)
reservoir storage block	res(j) (type hyd_output)
reservoir inflow and outflow blocks	res_in_d, res_out_d
crop_yld_aa MASS, C, N, P	pl_mass%yield_tot(ipl) divided by pcom%plcur(ipl)%harv_num
plant and weather nplt, pplnt	hpw_d%nplnt, hpw_d%pplnt, both copied from pl_mass%tot_com
plant and weather solarad, phubas0	hpw_d%solrad, hpw_d%phubase0